

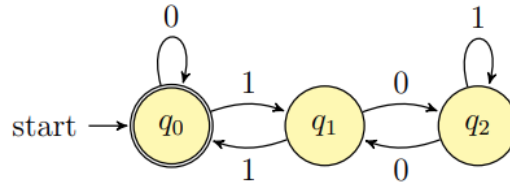
CSE331: Automata and Computability

Assignment 1

Deadline: 23/07/2026 11:59 PM

Question 1

Consider the finite automaton $(Q, \Sigma, \delta, q_0, F)$ depicted below



where $Q = \{q_0, q_1, q_2\}$, $\Sigma = \{0, 1\}$, and $F = \{q_0\}$.

- a) Find and fix the mistake in the following symbolic description of the transition function δ :

$Q \times \Sigma \rightarrow Q$: for each $j \in \{0, 1\}$

$$\delta(q_0, j) = q_j$$

$$\delta(q_1, j) = q_{1-j}$$

$$\delta(q_2, j) = q_{1+j}$$

The mistake is in $\delta(q_1, j)$. Correct description:

$$\delta(q_0, j) = q_j, \quad \delta(q_1, j) = q_{2-2j}, \quad \delta(q_2, j) = q_{1+j}$$

- b) Keeping the same set of states Q , alphabet Σ , starting state q_0 , and set of accepting states $\{q_0\}$, change the transition function δ so that the resulting finite automaton never accepts a string ending in a 1.

Change

$$\delta(q_1, 1) = q_1$$

Keep all other transitions unchanged. Then every input 1 leads to a non-accepting state, so no string ending in 1 is accepted.

- c) There is a beautiful plain English description of the language recognized by the finite automaton with the state diagram depicted at the start of Question 1. What is it?

$\{w \mid w \text{ when interpreted as a binary number, is divisible by } 3\}$

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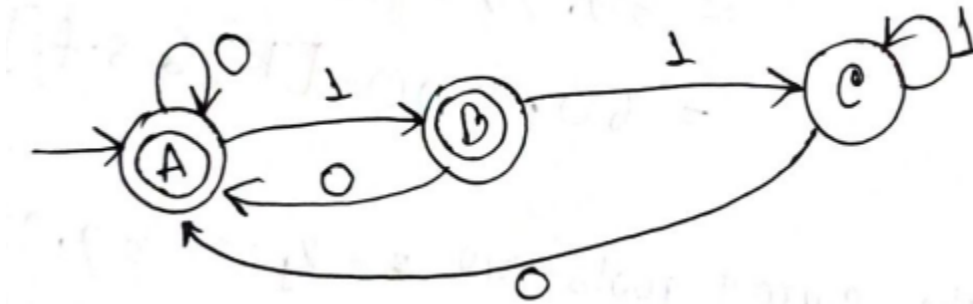
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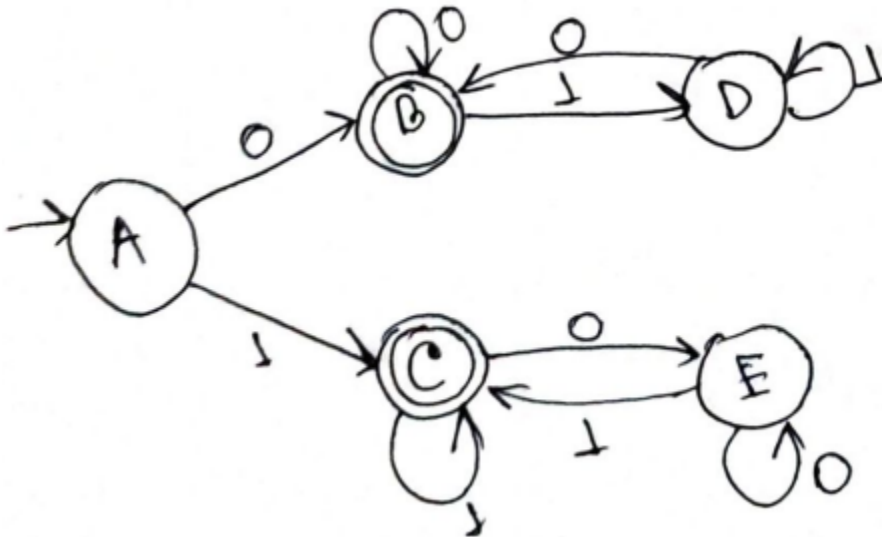
Question 2

For each of the languages below, draw the DFA that recognizes the language. You may assume that each language is over the alphabet $\Sigma = \{0, 1\}$.

- (a) $\{w \in \Sigma^* \mid \text{the set of all string which do not end with 11}\}$



- (b) $\{w \in \Sigma^* \mid \text{if the number of 1's in } w \text{ is odd, then the number of 0's in } w \text{ is also odd}\}$

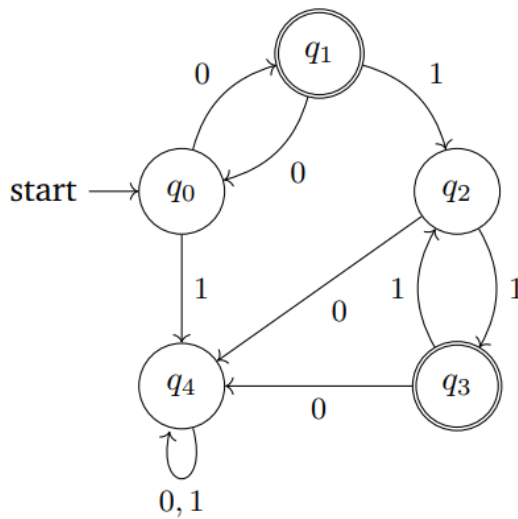


- (c) $\{w \in \Sigma^* \mid \text{binary strings that contain an even number of 1s and an odd number of 0's and do not contain the substring 10.}\}$ (You need to construct this language without cross product)

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Question 3

Let $\Sigma = \{a, b\}$. Consider the following languages over Σ .

$L_1 = \{w : \text{length of } w \text{ is two more than multiple of four}\}$

$L_2 = \{w : \text{every even position letter in } w \text{ is different from the first letter of } w\}$

$L_3 = \{w : \text{every } 2k + 1 \text{ position in } w \text{ is } b, \text{ where } k \geq 0\}$

$L_4 = \{w : \text{every } 2k + 1 \text{ position in } w \text{ is } a, \text{ where } k \geq 0\}$

Now solve the following problems.

- (a) Give the state diagram for a DFA that recognizes L_1 . (3 points)
- (b) Give the state diagram for a DFA that recognizes L_2 . (3 points)
- (c) Give the state diagram for a DFA that recognizes L_3 . (3 points)
- (d) If you were to use the “cross product” construction to obtain a DFA for the language $L_2 \cap (L_3 \cup L_4)$, how many states would it have? (1 point)
- (e) Find all four-letter strings in $L_2 \cap (L_3 \cup L_4)$. (1 point)
- (f) Give the state diagram for a DFA that recognizes $L_2 \cap (L_3 \cup L_4)$ using only four states. (2 points)
- (g) Find a four-letter string in $\overline{L_3} \circ L_4$. [Recall: $\overline{L}$ denotes the complement of the language L i.e., $\overline{L} = \Sigma^* - L$] (1 point)
- (h) Is $\overline{L_3} \circ L_4 = \overline{L_3}$? Give justification for your answer. (1 point)

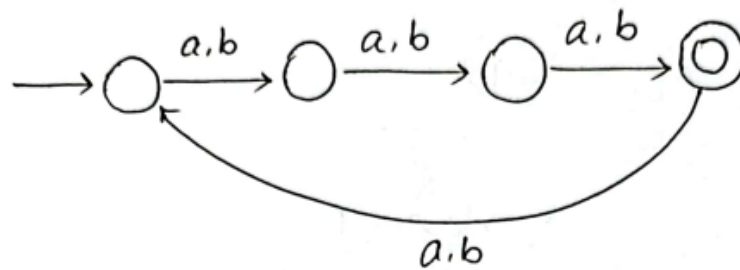
Solution:

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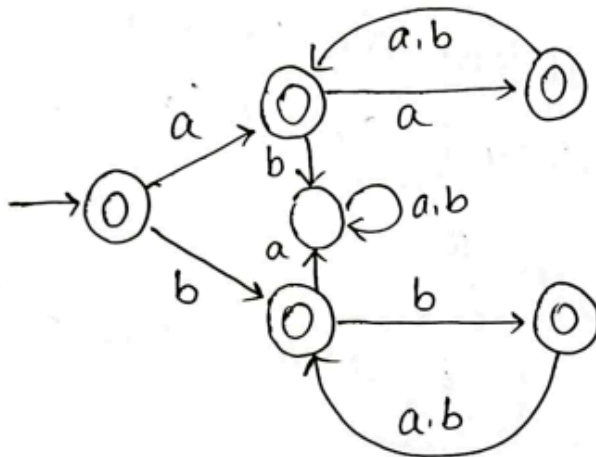
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a) $L1 = \{ \text{length of } w \text{ is three more than multiple of four.} \}$



b) $L2 = \{ \text{every even position letter in } w \text{ is the same as the first letter of } w \}$

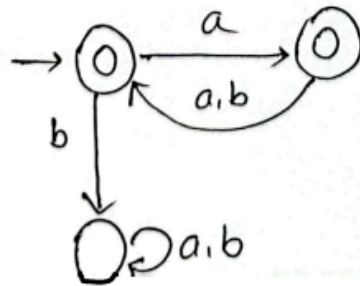


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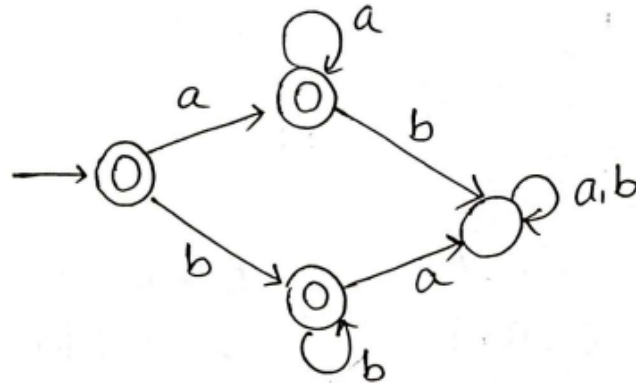
c) $L_3 = \{ \text{every } 2K+1 \text{ position in } w \text{ is } a, \text{ where } K \geq 0 \}$



d) $6 * 3 * 3 = 54$

e) aaaa, bbbb

f) $L_2 \cap (L_3 \cup L_4)$



g) $\overline{L_3} \cap L_4$ contains b in at least one $2k+1$ position in L_4 is b

'b'aaa, 'b'aab, 'b'aba, 'b'abb, 'b'baa, 'b'bab, 'b'bba, 'b'bbb, aa'b'a, aa'b'b, ab'b'a, ab'b'b

h) $\overline{L_3} \cap L_4 = \overline{L_3}$

Justification: Since, L_4 accepts Empty string, ϵ hence we can say $\overline{L_3} \cap L_4 = \overline{L_3}$.

Question 4

For each of the languages below, write the regular expression for the given language. You may assume that each language is over the alphabet $\Sigma = \{0, 1\}$.

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- (a) $\{w \in \Sigma^* \mid w \text{ contains equal number of 01 and 10}\}$
 $0\Sigma^*0 + 1\Sigma^*1 + 0 + 1 + \epsilon$
- (b) $\{w \in \Sigma^* \mid w \text{ do not contain substring ba}\}$
 a^*b^*
- (c) $\{w \in \Sigma^* \mid w \text{ is the set of all strings whose 1st,3rd,5th, } \dots, \text{ characters are the same}\}$
 $\epsilon + 1 + 0 + 1(\Sigma 1)^*(\Sigma + \epsilon) + 0(\Sigma 0)^*(\Sigma + \epsilon)$
- (d) $\{w \in \Sigma^* \mid w \text{ starts with 0 and has odd length, or starts with 1 and has even length}\}$
 $0(\Sigma\Sigma)^* + 1\Sigma(\Sigma\Sigma)^*$

Question 5

Let $\Sigma = \{0, 1\}$. Consider the following languages over Σ .

$$L_1 = \{w \text{ contains exactly two } 1\}$$

$$L_2 = \{w \text{ doesn't start with } 0\}$$

$$L_3 = \{w \text{ every third position in } w \text{ is } 1\}$$

$$L_4 = \{w \text{ every } 1 \text{ in } w \text{ is followed by at least two } 0\}$$

$$L_5 = L_3 \cap L_4$$

Now solve the following problems.

- (a) Give a regular expression for the language L_1 . (3 points)
- (b) Give a regular expression for the language L_2 . (3 points)
- (c) Give a regular expression for the language L_3 . (3 points)
- (d) Write a five-letter string that belongs to L_5 . (1 point)
- (e) Give a regular expression for the language L_5 . (2 points)
- (f) Give a regular expression for the language $\bar{L}_4$. Here $\bar{L}$ denotes the complement of the language L i.e., $\bar{L} = \Sigma^* - L$. (3 points)

(a) $0^*10^*10^*$

(b) $1(0+1)^* + \epsilon$

(c) $(\Sigma\Sigma 1)^*(\epsilon + \Sigma + \Sigma\Sigma)$

(d) 00100

(e) $\Sigma^*1(0+\epsilon)(1\Sigma^*+\epsilon)$

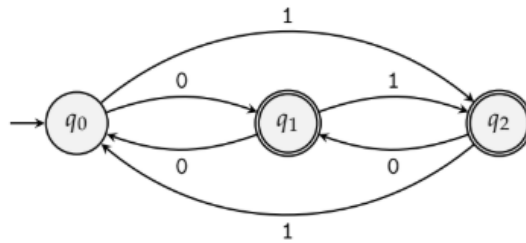
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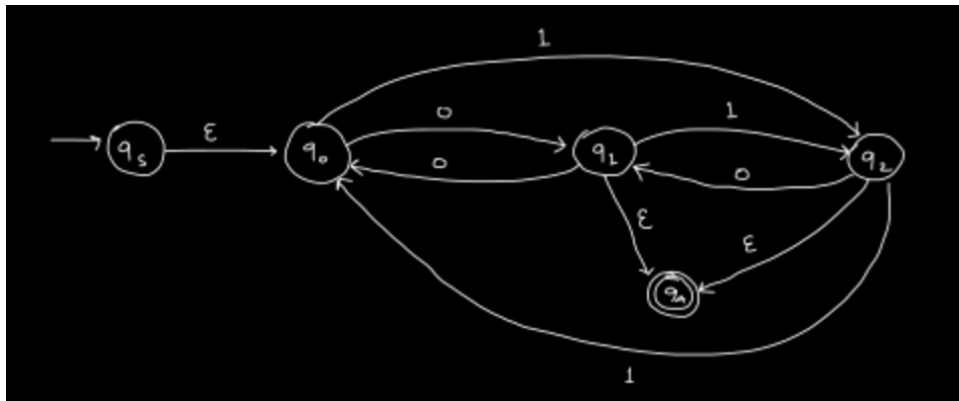
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Question 6

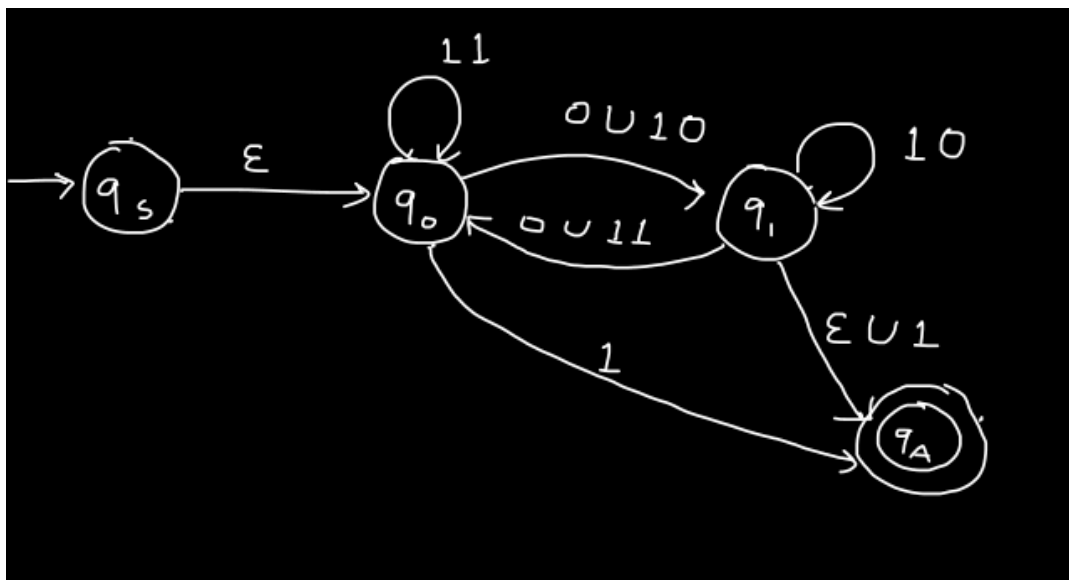
Convert the following DFA into an equivalent regular expression using the state elimination method. First eliminate q_2 , then q_1 , and finally q_0 . You must show work.



Adding new start and accepting states (1 point)



Eliminate q_2 (3 points)



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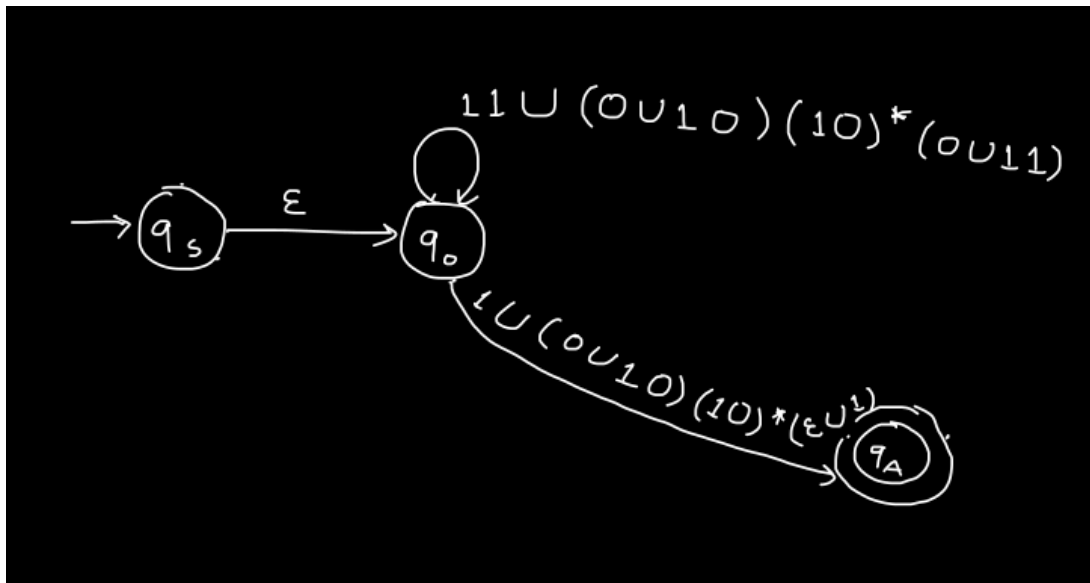
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- $q_0 \rightarrow q_0$ 0.5 points
- $q_0 \rightarrow q_1$ 0.5 points
- $q_0 \rightarrow q_A$ 0.5 points
- $q_1 \rightarrow q_0$ 0.5 points
- $q_1 \rightarrow q_1$ 0.5 points
- $q_1 \rightarrow q_A$ 0.5 points

Eliminate q_1 (4 points)

- $q_0 \rightarrow q_0$ 2 points
- $q_0 \rightarrow q_A$ 2 points



Eliminate q_0 (1 point)

